

Problem Set 4 Answers

1. On the first three cards, with the given condition that no Aces appear until the 4th draw, he is essentially drawing 3 cards (without replacement) from a deck of 48 cards, with all outcomes equally-likely. So the conditional probability that:

1a. there are 0 Kings is $\binom{3}{0}(44/48)(43/47)(42/46)$;

1b. there is 1 King is $\binom{3}{1}(4/48)(44/47)(43/46)$;

1c. there are 2 Kings is $\binom{3}{2}(4/48)(3/47)(44/46)$;

1d. there are 3 Kings is $\binom{3}{3}(4/48)(3/47)(2/46)$.

2. There are 23 equally-likely ways that the sum on the pair of dice is between 4 and 8 (inclusive). So the conditional probability that:

2a. the sum equals 4 is $3/23$;

2b. the sum equals 5 is $4/23$;

2c. the sum equals 6 is $5/23$;

2d. the sum equals 7 is $6/23$;

2e. the sum equals 8 is $5/23$.

3. There are $6^3 - 5^3 = 91$ equally-likely families with at least one green bear.

So the conditional probability that:

3a. the bear family has exactly 1 green bear is $\binom{3}{1}(5^2/91) = 75/91$;

3b. the bear family has exactly 2 green bears is $\binom{3}{2}(5/91) = 15/91$;

3c. the bear family has exactly 3 green bears is $\binom{3}{3}/91 = 1/91$.

4. There are 14 equally-likely ways that Bob's result is strictly smaller than Alice's result. So the conditional probability that:

4a. Bob's die shows the value 1 is $5/14$;

4b. Bob's die shows the value 2 is $4/14$;

4c. Bob's die shows the value 3 is $3/14$;

4d. Bob's die shows the value 4 is $2/14$.

5. There are 6 equally-likely ways that Alice's result is strictly smaller than Bob's result. So the conditional probability that:

5a. Alice's die shows the value 1 is $3/6$;

5b. Alice's die shows the value 2 is $2/6$;

5c. Alice's die shows the value 3 is $1/6$.